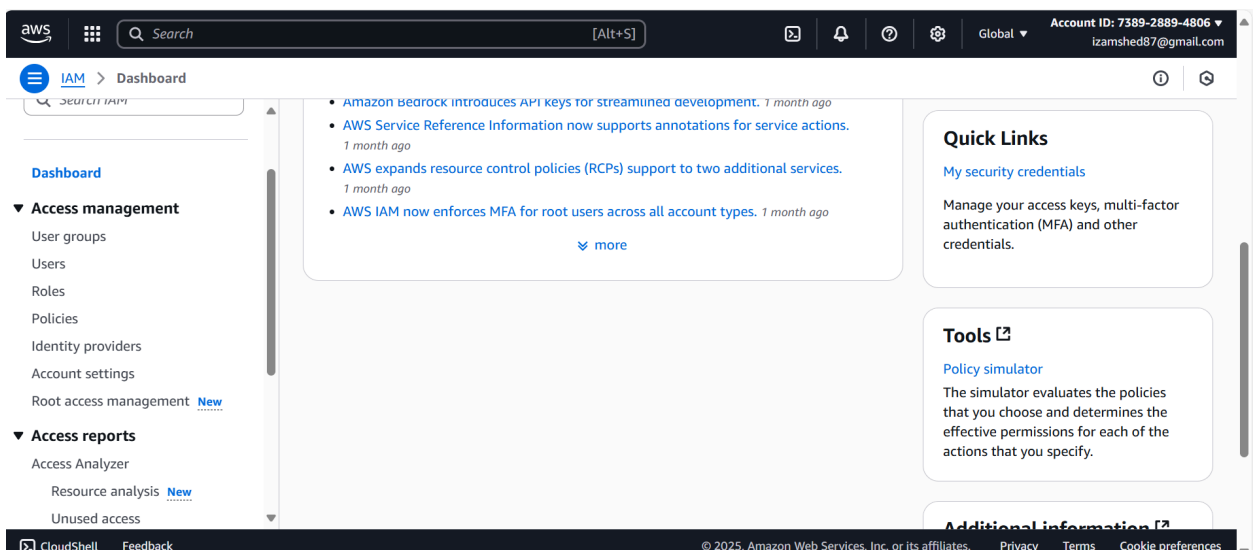


1.Project-Structure

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ ls
README.md  main.tf  outputs.tf  terraform.tfvars  variables.tf
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ pwd
/mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ |
```

2.Creating Access Key & Secret Key

IAM > Dashboard > Quick Link > Security Credentials > create access key



aws

Search

[Alt+S]

Global

Account ID: 7389-2889-4806
izamshed87@gmail.com

IAM > Security credentials > Create access key

Step 1
Access key best practices & alternatives

Step 2 - optional
Set description tag

Step 3
Retrieve access keys

Access key best practices & alternatives

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

Use case

☐ Command Line Interface (CLI)
You plan to use this access key to enable the AWS CLI to access your AWS account.

☐ Local code
You plan to use this access key to enable application code in a local development environment to access your AWS account.

☐ Application running on an AWS compute service
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.

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Account ID: 7389-2889-4806
izamshed87@gmail.com

IAM > Security credentials

Identity and Access Management (IAM)

eks-admin

Access keys (1)

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

AKIA2YC4YENLBCQLHKFI

Description

Access denied

Last used

You need permissions

Last used region

You need permissions

Status

Active

Created

Now

Last used service

You need permissions

Actions

CloudShell Feedback

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3. Terraform initialization, validation

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform -v
Terraform v1.12.2
on linux_amd64
+ provider registry.terraform.io/hashicorp/aws v6.9.0
+ provider registry.terraform.io/hashicorp/random v3.7.2
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/random from the dependency lock file
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.9.0
- Using previously-installed hashicorp/random v3.7.2

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform validate
Success! The configuration is valid.
```

4. Terraform Execution Plan preview

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform plan
data.aws_ami.amazon_linux2: Reading...
data.aws_ami.amazon_linux2: Read complete after 3s [id=ami-0e2c86481225d3c51]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform will perform the following actions:

# aws_instance.web will be created
+ resource "aws_instance" "web" {
  + ami              = "ami-0e2c86481225d3c51"
  + arn              = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone = (known after apply)
  + disable_api_stop  = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized     = (known after apply)
  + enable_primary_ipv6 = (known after apply)
  + force_destroy     = false
  + get_password_data = false
  + host_id           = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile = (known after apply)
  + id                = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle = (known after apply)
  + instance_state     = (known after apply)
  + instance_type      = "t2.micro"
  + ipv6_address_count = (known after apply)
  + ipv6_addresses     = (known after apply)
  + key_name           = (known after apply)
}
```

5.Terraform apply (Infra Creation)

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform apply --auto-approve
data.aws_ami.amazon_linux2: Reading...
data.aws_ami.amazon_linux2: Read complete after 3s [id=ami-0e2c86481225d3c51]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform will perform the following actions:

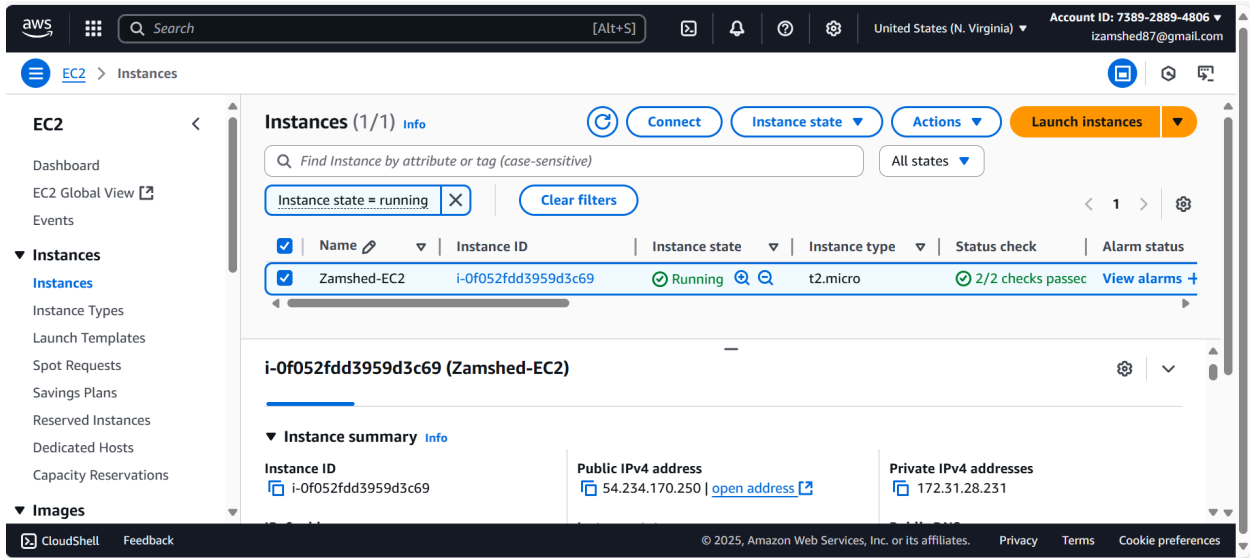
# aws_instance.web will be created
+ resource "aws_instance" "web" {
  + ami                  = "ami-0e2c86481225d3c51"
  + arn                  = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone     = (known after apply)
  + disable_api_stop      = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized         = (known after apply)
  + enable_primary_ipv6    = (known after apply)
  + force_destroy         = false
  + get_password_data      = false
  + host_id               = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile   = (known after apply)
  + id                    = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle     = (known after apply)
  + instance_state         = (known after apply)
  + instance_type          = "t2.micro"
  + ipv6_address_count     = (known after apply)
  + ipv6_addresses         = (known after apply)
  + key_name               = (known after apply)
}
```

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

```
aws_region = "us-east-1"
ec2_instance_id = "i-0f052fdd3959d3c69"
s3_bucket_name = "tf-assignment-9deb"
```

6.Output-EC2 & S3 in AWS

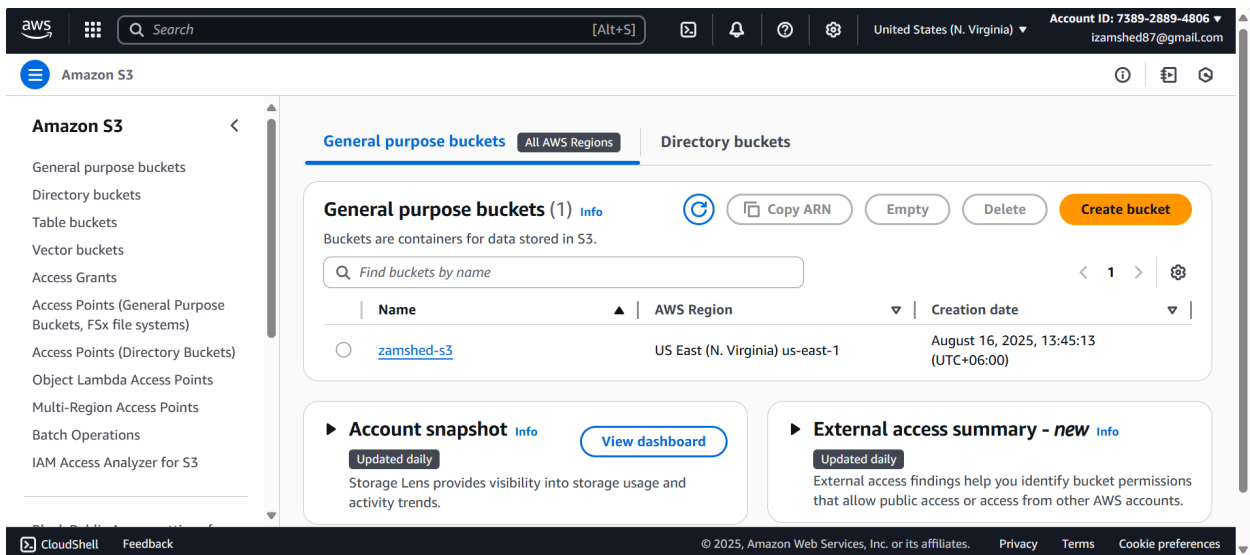


The screenshot displays the AWS Management Console for the EC2 service. The left sidebar shows the navigation menu with 'Instances' selected. The main content area shows a list of instances with one instance, 'Zamshed-EC2', in the 'Running' state. Below the list, the 'Instance summary' for 'i-0f052fdd3959d3c69' is shown, including its ID, public IPv4 address (54.234.170.250), and private IPv4 addresses (172.31.28.231).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
Zamshed-EC2	i-0f052fdd3959d3c69	Running	t2.micro	2/2 checks passed	View alarms

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0f052fdd3959d3c69	54.234.170.250 open address	172.31.28.231



The screenshot displays the AWS Management Console for the Amazon S3 service. The left sidebar shows the navigation menu with 'Amazon S3' selected. The main content area shows the 'General purpose buckets' tab, which lists one bucket, 'zamshed-s3', in the 'US East (N. Virginia) us-east-1' region, created on August 16, 2025, at 13:45:13 (UTC+06:00).

Name	AWS Region	Creation date
zamshed-s3	US East (N. Virginia) us-east-1	August 16, 2025, 13:45:13 (UTC+06:00)

Account snapshot

Storage Lens provides visibility into storage usage and activity trends.

External access summary - new

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

7.Terraform Destroy (Infra Deletion)

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform destroy
data.aws_ami.amazon_linux2: Reading...
aws_s3_bucket.bucket: Refreshing state... [id=zamshed-s3]
data.aws_ami.amazon_linux2: Read complete after 4s [id=ami-0e2c86481225d3c51]
aws_instance.web: Refreshing state... [id=i-0f052fdd3959d3c69]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
  - destroy

Terraform will perform the following actions:

# aws_instance.web will be destroyed
- resource "aws_instance" "web" {
  - ami              = "ami-0e2c86481225d3c51" -> null
  - arn              = "arn:aws:ec2:us-east-1:738928894806:instance/i-0f052fdd3959d3c69" -> null
  - associate_public_ip_address = true -> null
  - availability_zone = "us-east-1b" -> null
  - disable_api_stop    = false -> null
  - disable_api_termination = false -> null
  - ebs_optimized       = false -> null
  - force_destroy       = false -> null
  - get_password_data   = false -> null
  - hibernation         = false -> null
  - id                 = "i-0f052fdd3959d3c69" -> null
  - instance_initiated_shutdown_behavior = "stop" -> null
  - instance_state     = "running" -> null
  - instance_type      = "t2.micro" -> null
  - ipv6_address_count = 0 -> null
  - ipv6_addresses     = [] -> null
  - monitoring         = false -> null
  - placement_partition_number = 0 -> null
  - primary_network_interface_id = "eni-0a5c6e3905a14a763" -> null
}
```

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_s3_bucket.bucket: Destroying... [id=zamshed-s3]
aws_instance.web: Destroying... [id=i-0f052fdd3959d3c69]
aws_s3_bucket.bucket: Destruction complete after 2s
aws_instance.web: Still destroying... [id=i-0f052fdd3959d3c69, 00m10s elapsed]
aws_instance.web: Still destroying... [id=i-0f052fdd3959d3c69, 00m20s elapsed]
aws_instance.web: Destruction complete after 25s
```

Destroy complete! Resources: 2 destroyed.

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$
```